Leah English

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Research Interest

Computational Astrophysics - galactic dynamics, dark matter probes, underlying dark matter models

Education

B.S. in Physics **- Illinois Institute of Technology** Chicago, IL **May 2025**
B.S. in Astrophysics **- Illinois Institute of Technology** Chicago, IL **May 2025**

Publications, Presentations, & Papers in Progress

1. Cook, B. T., Tep, K., Rodriguez, C.L., **English, L.**, Starkenburg, T., Sanderson, R., Weatherford, N., Pearson, S. & Panithanpaisal, N. (2026). "Modeling Globular Cluster Stellar Streams with a Basis-Expansion N-Body Code." *The Astrophysical Journal, 1010(2), Article 155*. https://doi.org/10.3847/1538-4357/aea6a2
Contribution: Modeled stellar streams using particle-spray models to compare those produced by the developing N-body code, KRIOS
2. **English, L**., Starkenburg, T., Weerasooriya, S. "Dancing Streams in Merging Halos: Stellar Streams Accreted from Satellite Galaxies" *Paper in preparation*.
Contribution: Analyzing modeled stellar streams in a satellite galaxy as it merges with a host galaxy. Comparing streams within the host galaxy as well as observed stellar streams.
3. **English, L**., Leiner, Emily. "The Trajectory of WOCS 5379: a Blue Straggler-White Dwarf Binary" *Research note in preparation*.
Contribution: Used MESA to model the future trajectory of WOCS 5379, an observed BSS-WD binary, exploring the effects of different mass transfer prescriptions
4. **English, L.**, Starkenburg, T., Weerasooriya, S. "Dancing Stellar Streams in Merging Dark Matter Halos" *Poster Presentation at DDA 57*; June 2026; Chicago, IL
5. **English, L**., Starkenburg, T., Weerasooriya, S. "Unravelling the Dynamics of Stellar Streams in Merging Galaxies" *Poster Presentation at AAS 245*; January 2025; National Harbor, MD
Chambliss student finalist
6. Hussain, A., **English, L.**, Leiner, E. "The Effect of Common Envelope Efficiency and Metallicity on Rate of Blue Straggler Binary Mergers" *Poster Presentation at AAS 243*; January 2024; New Orleans, LA
Contribution: Created models for a case study of WOCS 5379
7. **English, L**., Leiner, E., Hussain, H. "Modeling the Future Evolution of a Blue Straggler Star-White Dwarf Binary System" *Poster Presentation at the Illinois Tech Summer Research Exposition*; August 2023; Chicago, IL
Finalist

Research Experience

Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA) - Northwestern University Evanston, IL
Assistant Researcher **September 2024 - Present**

- Numerically model stellar streams in a satellite galaxy merging with a host galaxy using a C++ based cosmological code, GADGET-4, and Northwestern's Quest High-Performance Computing Cluster supercomputer. Explore effects of mass, orbit, and number of stars as they merge. Visualization and analysis using Pandas and Numpy in Python
- Model stellar streams using particle spray methods in Python's Gala package to compare to streams produced by KRIOS, a developing N-Body code at UNC-Chapel Hill. Analyze streams using 2D histograms and density mapping.

Research Experience for Undergraduates (REU) **June 2024 - August 2024**

- Analyze large sets of data from numerical simulations of stellar streams using Python. Visualize streams through movies and calculate their energies, angular momentum, and orbital coordinates using Python
- Involvement in Python workshops, presentation workshops, and GitHub workshops

Illinois Institute of Technology Chicago, IL
Student Research Assistant (NASA Illinois Space Grant) **June 2023 - May 2025**

- Model the evolution of a binary star system using Fortran-based 1D stellar evolutionary code MESA, exploring effects of mass transfer prescriptions. Analyze and visualize data using Python

Technical Skills

Programming: Python, C++, Fortran
Scientific Computing and Data Analysis: NumPy, SciPy, Pandas, Matplotlib, numerical simulation analysis, statistical comparison, scientific data visualization
Astrophysics Software and Computing: GADGET-4, Gala, high-performance computing (Northwestern QUEST cluster), MESA
Research Tools: GitHub, LaTeX

Additional Professional Experience

SpiderRock, Chicago IL - *Market Data Entitlement Administrator* **August 2025 - Present**

- Review and improve operational tasks through automation to increase efficiency and reduce manual errors
- Liaison between internal departments and external data vendors
- Ensure compliance with data licensing rules and regulations
- Manage and control user access to market data, ensuring compliance with vendor licensing agreements
- Maintain accurate inventory of data, contracts, services, and users

Just Roots, Chicago IL - *Growing Young Leader Intern* **June 2025 - August 2025**

- Support farm operations and practice regenerative growing, providing food to the community
- Engage in a variety of workshops that address health and wellness, money management, and other holistic life skills.

Leadership & Outreach

Illinois Tech Admissions Office, Chicago, IL
Hawk Ambassador **August 2022 - May 2025**

- Tour guide, answer calls and emails, point of contact at prospective students. Create a friendly environment to encourage visitors to come back.

Senior Lead Hawk Ambassador **January 2024 - October 2024**

- Recruit, interview, and train Hawk Ambassadors, create schedules, manage progress in the office. Run monthly meetings.
- Stepped down to focus on research

Illinois Tech Physics Department, Chicago, IL
Physics Outreach Ambassador **January 2024 - May 2024**
Advise accepted physics and astrophysics students to aid their transition to Illinois Tech.

- Join Zoom meetings to answer any questions they may have about the program
- Run the Illinois Tech Physics Instagram page to promote the department